DSA Interview Preparation

Complete 100 Problems Checklist | Topics: String, Sliding Window, Queue, Recursion, Backtracking, Greedy, Graph, DP, 2D Matrix

1. STRING (25 problems)

■ **Done**

■ Anagram Check
■ Find / Remove Banned Words
■ Check if Subsequence of Another String
■ Compress String
■ String Compression II
■ Backspace String Compare
■ Longest Substring Without Repeating Characters
■ Extract Numbers and Find Largest
■ Partition Labels

■ **To Do**

■ Reverse Words in a String
■ Valid Palindrome
■ Valid Palindrome II (one deletion allowed)
■ Longest Palindromic Substring
■ Roman to Integer / Integer to Roman
■ Count and Say
■ Minimum Window Substring
■ Find All Anagrams in a String
■ Permutation in String
■ Repeated Substring Pattern
■ String to Integer (atoi)
■ Longest Common Prefix
■ Group Anagrams
■ Decode String (Stack based)
■ Word Break
■ Wildcard Matching / Regular Expression (Hard)

2. SLIDING WINDOW (9 problems — DONE)

■ **Done**

■ Longest Substring Without Repeating Characters
■ Maximum Sum of Distinct Subarrays With Length K
■ Maximum Average Subarray I
■ First Negative Number in Every Window of Size K
■ First Smallest Number in Every Window Size K (Frequency Array)
■ Sliding Subarray Beauty (Frequency Array)
■ Subarrays with K Different Integers (Two Pointer + Sliding Window)
■ Sliding Window Maximum (Hard)
■ Minimum Window Substring (Variable Window — High Priority)

3. QUEUE / DEQUE (10 problems)

■ **Done**

■ Sliding Window Maximum
■ First Negative in Every Window

■ **To Do**

■ Sliding Window Minimum
■ Shortest Subarray with Sum at Least K
■ Implement Queue using Stacks

■ Implement Stack using Queue
■ Binary Tree Level Order Traversal
■ Rotting Oranges
■ Design Hit Counter / Number of Recent Calls
■ Gas Station

4. RECURSION (5 problems)

■ To Do

■ Factorial / Fibonacci
■ Power of X (Fast Exponentiation)
■ Reverse a String using Recursion
■ Sum of Digits
■ Tower of Hanoi

5. BACKTRACKING (10 problems)

■ To Do

■ Subsets
■ Subsets II (with duplicates)
■ Permutations I
■ Permutations II
■ Combination Sum I
■ Combination Sum II
■ Letter Combinations of Phone Number
■ Palindrome Partitioning
■ N-Queens (Hard)
■ Sudoku Solver (Hard)

6. GREEDY (13 problems)

■ To Do

■ Coin Change (Greedy — standard denominations)
■ Activity Selection Problem
■ Fractional Knapsack
■ Job Sequencing Problem
■ Minimum Number of Platforms
■ Jump Game I
■ Jump Game II
■ Gas Station
■ Candy Distribution (Hard)
■ Meeting Rooms I
■ Meeting Rooms II
■ Merge Intervals
■ Non Overlapping Intervals

7. GRAPH (12 problems)

■ To Do

■ Number of Islands (BFS/DFS)
■ Clone Graph
■ Flood Fill
■ Rotting Oranges
■ Course Schedule I — Cycle Detection
■ Course Schedule II — Topological Sort
■ Pacific Atlantic Water Flow
■ Network Delay Time — Dijkstra
■ Min Cost to Connect Points — MST

■ Bipartite Check
■ Word Ladder
■ Number of Provinces

8. DYNAMIC PROGRAMMING (20 problems)

■ To Do — 1D DP

■ Climbing Stairs
■ House Robber I
■ House Robber II
■ Min Cost Climbing Stairs
■ Decode Ways
■ Maximum Subarray — Kadane's Algorithm

■ To Do — String DP

■ Longest Common Subsequence
■ Longest Palindromic Substring
■ Edit Distance
■ Word Break

■ To Do — Knapsack

■ 0/1 Knapsack
■ Partition Equal Subset Sum
■ Target Sum
■ Coin Change I — Min Coins
■ Coin Change II — Number of Ways

■ To Do — 2D DP

■ Unique Paths
■ Minimum Path Sum
■ Maximal Square

■ To Do — Hard

■ Burst Balloons
■ Longest Increasing Subsequence

9. 2D MATRIX (8 problems)

■ To Do

■ Spiral Matrix
■ Rotate Image
■ Set Matrix Zeroes
■ Search a 2D Matrix
■ Word Search (Backtracking + Matrix)
■ Number of Islands
■ Surrounded Regions
■ Game of Life

SUMMARY

Topic	Total	Done	Remaining
String	25	9	16
Sliding Window	9	9	0
Queue / Deque	10	2	8
Recursion	5	0	5

Backtracking	10	0	10
Greedy	13	0	13
Graph	12	0	12
Dynamic Programming	20	0	20
2D Matrix	8	0	8
TOTAL	**112**	**20**	**92**

Recommended Order: Recursion → Backtracking → Greedy → Graph → DP → 2D Matrix

Target: ~3-4 problems/day | Estimated time to complete: 6-7 weeks